

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404015
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	M N; Hemanth et al.

Energy absorption system for landing gear

Abstract

A shock strut assembly is disclosed herein. The shock strut assembly includes a mixed fluid chamber, a liquid chamber coupled to the mixed fluid chamber, a metering pin having a first end and a second end and a channel formed therein extending from the first end to the second end, the first end disposed in the liquid chamber and the second disposed in the mixed fluid chamber, and a pressure relief valve fluidly coupled to the liquid chamber and the channel in the metering pin, the pressure relief valve configured to allow a flow of a fluid from the liquid chamber to the channel in the metering pin.

Inventors:	M N; Hemanth (Hassan, IN), Bodki; Basavaraj (Bangalore, IN)
Applicant:	Goodrich Corporation (Charlotte, NC)
Family ID:	93745312
Assignee:	Goodrich Corporation (Charlotte, NC)
Appl. No.:	18/447935
Filed:	August 10, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240409207 A1	Dec. 12, 2024

Foreign Application Priority Data

IN	202341039006	Jun. 07, 2023
----	--------------	---------------

Publication Classification

Int. Cl.:	B64C25/60 (20060101)
------------------	----------------------

U.S. Cl.:

CPC **B64C25/60** (20130101);

Field of Classification Search

CPC: B64C (25/60); F16F (9/342); F16F (9/486)

USPC: 267/64.18

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3073586	12/1962	Hartel	N/A	N/A
3147826	12/1963	McHenry	188/289	B60G 15/12
3152667	12/1963	Powell	188/289	F16F 9/486
3458016	12/1968	Keech	N/A	N/A
3696894	12/1971	Brady	188/289	F16F 9/486
4054311	12/1976	Gute	267/116	B60R 19/32
4787486	12/1987	Hrusch et al.	N/A	N/A
5598903	12/1996	Richardson	188/269	F16F 9/504

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0262246	12/1985	EP	N/A
992347	12/1950	FR	N/A

OTHER PUBLICATIONS

European Patent Office, European Search Report dated Oct. 31, 2024 in Application No. 24180011.9. cited by applicant

Primary Examiner: Abell; Tye William

Attorney, Agent or Firm: SNELL & WILMER L.L.P.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to, and the benefit of, India Provisional Patent Application No. 202341039006, filed Jun. 7, 2023 (DAS Code 2D4F) and titled “ENERGY ABSORPTION SYSTEM FOR LANDING GEAR,” which is incorporated by reference herein in its entirety for all purposes.

FIELD

(2) The present disclosure generally relates to aircraft landing systems and, more particularly, to systems and methods to improve shock strut performance

BACKGROUND

(3) Shock absorbing devices are used in a wide variety of vehicle suspension systems for

controlling motion of the vehicle and its tires with respect to the ground and for reducing transmission of transient forces from the ground to the vehicle. Shock absorbing struts are a common component in most aircraft landing gear assemblies. The shock struts used in the landing gear of aircraft generally are subject to more demanding performance specifications than many ground vehicle shock absorbers. In particular, shock struts control motion of the landing gear, and absorb and dampen loads imposed on the landing gear during braking, landing, taxiing, and takeoff.

(4) A shock strut generally accomplishes these functions by compressing a fluid within a sealed chamber formed by hollow telescoping cylinders. The fluid generally includes both a gas and a liquid, such as hydraulic fluid or oil. One type of shock strut generally utilizes an “air-over-oil” arrangement wherein a trapped volume of gas is compressed as the shock strut is axially compressed, and a volume of oil is metered through an orifice. The gas acts as an energy storage device, such as a spring, so that upon termination of a compressing force the shock strut returns to its original length. Shock struts also dissipate energy by passing the oil through the orifice so that as the shock absorber is compressed or extended, its rate of motion is limited by the damping action from the interaction of the orifice and the oil.

(5) While shock struts may effectively absorb energy during aircraft landing events, “hard-landing events” may result in structural damage to the aircraft and/or landing gear assemblies. A landing event is conventionally deemed “hard” (i.e., a hard-landing event) when there is a spike in pressure in the shock strut that exceeds an upper design pressure threshold. This may also be referred to as an overloading event.

SUMMARY

(6) A shock strut assembly is disclosed herein. The shock strut assembly includes a mixed fluid chamber, a liquid chamber coupled to the mixed fluid chamber, a metering pin having a first end and a second end and a channel formed therein extending from the first end to the second end, the first end disposed in the liquid chamber and the second disposed in the mixed fluid chamber, and a pressure relief valve fluidly coupled to the liquid chamber and the channel in the metering pin, the pressure relief valve configured to allow a flow of a fluid from the liquid chamber to the channel in the metering pin.

(7) In various embodiments, the shock strut assembly further includes a circulation tube fluidly coupled to the pressure relief valve and a joint fluidly coupled to the channel and the circulation tube. In various embodiments, the pressure relief valve is configured to allow the flow of the fluid in response to a pressure in the shock strut assembly being above a threshold. In various embodiments, the pressure relief valve is configured to prevent the flow of the fluid in response to the pressure in the shock strut assembly being below the threshold.

(8) In various embodiments, the pressure relief valve is configured to prevent the flow of the fluid from the channel to the liquid chamber. In various embodiments, the second end of the metering pin includes an opening to the channel along an axis of the metering pin. In various embodiments, the second end of the metering pin includes a first opening to the channel and a second opening to the channel that are perpendicular to an axis of the metering pin.

(9) Also disclosed herein is a shock strut assembly including a shock strut cylinder defining a mixed fluid chamber, a shock strut piston defining a liquid chamber, the shock strut piston operatively coupled to slide within the shock strut cylinder, a metering pin having a first end and a second end and configured to extend from the shock strut piston into the shock strut cylinder, a channel formed within the metering pin from the first end to the second end, the channel configured to allow a flow of a liquid from the liquid chamber to the mixed fluid chamber, and a pressure relief valve fluidly coupled to the liquid chamber and the channel.

(10) In various embodiments, the shock strut assembly further includes a circulation tube fluidly coupled to the pressure relief valve and a joint fluidly coupled to the channel and the circulation tube. In various embodiments, the pressure relief valve is configured to allow the flow of the liquid in response to a pressure in the shock strut assembly being above a threshold. In various

embodiments, the pressure relief valve is configured to prevent the flow of the liquid in response to a pressure in the shock strut assembly being below a threshold. In various embodiments, the pressure relief valve is configured to prevent the flow of the liquid from the channel to the liquid chamber.

(11) In various embodiments, the shock strut piston is configured to slide in a first direction and wherein the second end of the metering pin includes an opening to the channel that is along the first direction. In various embodiments, the shock strut piston is configured to slide in a first direction and the second end of the metering pin includes a first opening extending a second direction through a sidewall of the metering pin, the second direction being perpendicular to the first direction.

(12) Also disclosed herein is a landing gear assembly including a shock strut, a wheel assembly coupled to the shock strut, and an energy absorption system coupled to the shock strut. The energy absorption system includes a shock strut cylinder defining a mixed fluid chamber, a shock strut piston defining a liquid chamber, the shock strut piston operatively coupled to slide within the shock strut cylinder, a metering pin having a first end and a second end, the first end disposed in the shock strut piston and the second end disposed in the shock strut cylinder, a channel formed along an inner diameter of the metering pin from the first end to the second end, and a pressure relief valve coupled to the liquid chamber and the channel.

(13) In various embodiments, the energy absorption system further includes a circulation tube fluidly coupled to the pressure relief valve and a joint fluidly coupled to the channel and the circulation tube. In various embodiments, the pressure relief valve is configured to allow a flow of a fluid in response to a pressure in the shock strut being above a threshold. In various embodiments, the pressure relief valve is configured to prevent a flow of a fluid in response to a pressure in the shock strut being below a threshold. In various embodiments, the pressure relief valve is configured to prevent a flow of a fluid from the channel to the liquid chamber. In various embodiments, the second end of the metering pin includes a first opening to the channel and a second opening to the channel that are perpendicular to an axis of the metering pin.

(14) The foregoing features and elements may be combined in any combination, without exclusivity, unless expressly indicated herein otherwise. These features and elements as well as the operation of the disclosed embodiments will become more apparent in light of the following description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the following detailed description and claims in connection with the following drawings. While the drawings illustrate various embodiments employing the principles described herein, the drawings do not limit the scope of the claims.

(2) FIG. 1 illustrates an exemplary aircraft having landing gear, in accordance with various embodiments.

(3) FIGS. 2A and 2B illustrate a shock strut of an aircraft landing gear having a pressure within operating range, in accordance with various embodiments.

(4) FIGS. 3A and 3B illustrate a shock strut of an aircraft landing gear having a pressure higher than operating range, in accordance with various embodiments.

(5) FIGS. 4A and 4B illustrate different designs for metering pins for use in a shock strut of an aircraft landing gear, in accordance with various embodiments.

(6) FIG. 5 illustrates a plot of a shock strut load as a function of the shock strut stroke, in accordance with various embodiments.

DETAILED DESCRIPTION

(7) The following detailed description of various embodiments herein makes reference to the accompanying drawings, which show various embodiments by way of illustration. While these various embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that changes may be made without departing from the scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation. While these exemplary embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, it should be understood that other embodiments may be realized and that logical, chemical and mechanical changes may be made without departing from the spirit and scope of the invention. For example, the steps recited in any of the method or process descriptions may be executed in any order and are not necessarily limited to the order presented. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected, or the like may include permanent, removable, temporary, partial, full or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact. It should also be understood that unless specifically stated otherwise, references to “a,” “an” or “the” may include one or more than one and that reference to an item in the singular may also include the item in the plural. Further, all ranges may include upper and lower values and all ranges and ratio limits disclosed herein may be combined.

(8) Disclosed herein is a shock strut for an aircraft including one or more pressure relief valves to provide extra dampening during a hard landing, or overload, event. The one or more pressure relief valves allow additional fluid flow from a lower chamber in the shock strut to an upper chamber in the shock strut. In various embodiments, during an overload, or hard landing, event, the pressure inside the shock strut's lower chamber may increase drastically from a normal operating pressure range **P1** to an overloaded pressure range **P2** that is greater than the normal operating pressure range **P1**. In various embodiments, the one or more pressure relief valves may open in response to the pressure exceeding **P1**, allowing additional fluid to flow from the lower chamber to the upper chamber through an opening in a metering pin. In various embodiments, the one or more pressure relief valves may be unidirectional so that the fluid does not flow from the upper chamber to the lower chamber.

(9) Referring now to FIG. 1, an aircraft **100** is illustrated, in accordance with various embodiments. In accordance with various embodiments, aircraft **100** may include one or more landing gear assemblies, such as, for example, a left landing gear assembly **102** (or port-side landing gear assembly), a right landing gear assembly **104** (or starboard-side landing gear assembly) and a nose landing gear assembly **106**. Each of left landing gear assembly **102**, right landing gear assembly **104**, and nose landing gear assembly **106** may support aircraft **100** when not flying, allowing aircraft **100** to taxi, takeoff, and land safely and without damage to aircraft **100**. In various embodiments, left landing gear assembly **102** may include a left shock strut assembly **108** and a left wheel assembly **110**, right landing gear assembly **104** may include a right shock strut assembly **112** and a right wheel assembly **114**, and nose landing gear assembly **106** may include a nose shock strut assembly **116** and a nose wheel assembly **118**.

(10) Referring now to FIGS. 2A and 2B, a landing gear assembly **200**, similar to the landing gear assemblies described above with reference to FIG. 1, is illustrated. Landing gear assembly **200** includes an energy absorption system **250**. FIG. 2A illustrates a cross section of landing gear assembly **200** and FIG. 2B illustrates an enlarged cross section of energy absorption system **250** of landing gear assembly **200**. In accordance with various embodiments, landing gear assembly **200**

includes a shock strut **202**, which itself may include a shock strut cylinder **204** and a shock strut piston **206**. Shock strut piston **206** is operatively coupled to slide or telescope within shock strut cylinder **204**. Shock strut piston **206** may include a first end **208** disposed within shock strut cylinder **204** and a second end **210** extending from shock strut cylinder **204**. Shock strut cylinder **204** may be configured to receive shock strut piston **206** in a manner that allows the two components to telescope together and absorb and dampen forces transmitted thereto. FIGS. 2A and 2B illustrate shock strut **202** under a normal operating pressure **P1**. When in a compressed position, shock strut piston **206** will have translated, or slid, a distance (e.g., in the positive z-direction) into shock strut cylinder **204** from that illustrated in FIG. 2A.

(11) Shock strut cylinder **204** may be divided into a liquid chamber **212** and a mixed fluid chamber **214** as indicated by a line **213**. In various embodiments, a liquid, such as, for example, hydraulic fluid or oil may be located within liquid chamber **212**. Further, a gas, such as nitrogen or air, may be located within mixed fluid chamber **214**. In various embodiments, shock strut **202** includes an orifice plate **220** configured to separate liquid chamber **212** (also referred to herein as a first chamber or a dynamic liquid chamber) from mixed fluid chamber **214** (also referred to herein as a second chamber or a mixed gas/liquid chamber). In this regard, during operation, the liquid, such as, for example, hydraulic fluid or oil, may be located within liquid chamber **212** (or the first or dynamic liquid chamber) and a mixture of a gas, such as, for example, nitrogen or air, and the liquid may be located within mixed fluid chamber **214** (or the second or mixed gas/liquid chamber).

(12) Shock strut **202** includes a metering pin **222** coupled to shock strut piston **206** and configured to translate with shock strut piston **206**. In various embodiments, metering pin **222** is received by and extends through orifice plate **220**. Orifice plate **220** may comprise a metering pin aperture **224** configured to receive metering pin **222**. In various embodiments, metering pin **222** includes a metering flute (or a plurality of metering flutes) disposed along the length (e.g., along the Z-direction) of the metering pin **222**. As described herein, the metering flute defines a flow channel between metering pin **222** and metering pin aperture **224** of orifice plate **220** through which liquid may travel from liquid chamber **212** to mixed fluid chamber **214** in response to shock strut **202** moving to a compressed position from an extended position. Metering pin **222** has a first end **230** (e.g., the negative z-direction) and a second end **232** (e.g., the positive z-direction). In various embodiments, metering pin **222** may be tapered such that first end **230** being wider than second end **232**.

(13) In various embodiments, orifice plate **220** is housed within a retainer housing **226** and may include a retainer plate. Both orifice plate **220** and the retainer plate are typically annular in shape and secured to an orifice support tube **228**. Retainer housing **226** and, in various embodiments the retainer plate, are configured to restrict, but not prevent, the flow of fluid (e.g., liquid and/or gas) from liquid chamber **212** into mixed fluid chamber **214** during compression of shock strut **202**. Orifice plate **220** may translate away from retainer housing **226** during compression of shock strut **202** and translate toward retainer housing **226** during extension of shock strut **202**. In various embodiments, orifice plate **220** may be free to move within a channel formed in the orifice retainer plate.

(14) Referring now to FIG. 2B, the enlarged view of energy absorption system **250** further shows that the energy absorption system **250** includes one or more pressure relieve valves **252**, a circulation tube **254**, a joint **256**, a channel **258** formed through metering pin **222**, and an opening **260** formed in second end **232** of metering pin **222**. Energy absorption system **250** is configured to reduce the pressure in shock strut **202** in response to a pressure spike above normal operating pressure **P1** such as during hard landing event. The pressure in shock strut **202** is reduced by allowing a flow of the fluid in liquid chamber **212** to pass through the one or more pressure relieve valves **252**, through circulation tube **254**, joint **256**, channel **258**, and opening **260** into mixed fluid chamber **214**.

(15) In the illustrated embodiments, there are two pressure relieve valves **252**. In various embodiments, there may be one or more than two pressure relieve valves **252** depending on the aircraft and landing gear design. Pressure relief valves **252** are coupled to a bottom portion of liquid chamber **212** (e.g., in the z-direction) to allow fluid to leave liquid chamber **212**. In various embodiments, pressure relieve valves **252** may be unidirectional, allowing fluid to flow in a first direction but not in an opposite, second direction. That is, pressure relief valve **252** may allow fluid to flow out liquid chamber **212** through metering pin **222** (e.g., in the positive z-direction) but not into liquid chamber **212** through metering pin **222** (e.g., in the negative z-direction). In other words, pressure relief valve **252** may allow fluid to flow downward from liquid chamber **212** into circulation tube **254** (e.g., in the negative z-direction) but not upward from circulation tube **254** into liquid chamber **212** (e.g., in the positive z-direction)

(16) Circulation tube **254** may include a plurality of tubes, each of which is coupled to one pressure relief valve **252** at one end and to joint **256** at the other end. Circulation tube **254** provides a fluid path from pressure relief valve **252** to joint **256**.

(17) Joint **256** is coupled to metering pin **222**, and more specifically to channel **258**, at one end and to circulation tube **254** at the other end. In various embodiments, joint **256** may be a T-joint, as illustrated in FIGS. **2A** and **2B**, having two connection points for circulation tube **254**. In various embodiments, joint **256** may have more than two connection points for circulation tube **254**, depending on the number of pressure relief valves **252** in shock strut **202**.

(18) Channel **258** is formed along an inner diameter of metering pin **222** (e.g., along the z-axis). Channel **258** extends the entire length of metering pin **222** being coupled to joint **256** at one end and opening to mixed fluid chamber **214** through opening **260** at the other end. Channel **258** has a diameter that may vary based on design choices and an amount of fluid that will flow through channel **258**. In various embodiments, channel **258** includes a plurality of individual and/or separate channels formed through metering pin **222**. Therefore, it should be appreciated that while a single channel **258** is illustrated for ease of discussion and description, there may be, it should be appreciated that channel **258** may include a plurality of individual channels formed through metering pin **222**. Accordingly, in various embodiments, there may be multiple channels that may each extend through the entire length of metering pin **222**, may extend through a portion of the length of metering pin **222**, may split from a single channel into multiple channels, and may combine from multiple channels into a single channel, among other configurations, depending on design parameters.

(19) Referring now to FIGS. **3A** and **3B**, landing gear assembly **200** is illustrated under during a high pressure event, in accordance with various embodiments. FIG. **3B** illustrates a cross section of landing gear assembly **200** and FIG. **2B** illustrates an enlarged cross section of energy absorption system **250** of landing gear assembly **200**. FIGS. **3A** and **3B** illustrate shock strut **202** under an overload pressure **P2**.

(20) Generally, pressure relief valves **252** restrict a flow of liquid from liquid chamber **212** into circulation tube under normal operating pressures **P1**, as illustrated in FIGS. **2A** and **2B**. In various embodiments, normal operating pressure **P1** may be a range of pressures for which shock strut **202** is designed, as described in FIG. **5**. However, pressure relief valves **252** allow the flow of liquid into circulation tubes **254** in response to an overload pressure **P2**, as illustrated in FIGS. **3A** and **3B**, that is greater than normal operating pressure **P1**. Overload pressure **P2** may be any pressure that exceeds the upper range of normal operating pressure **P1**. In various embodiments, pressure relief valve **252** may be calibrated for use with each configuration of aircraft **100** and landing gear assembly **200** resulting in different ranges for normal operating pressure **P1** and overload pressure **P2** for each different configuration.

(21) During a hard landing, or overload, event, the pressure in shock strut **202** may increase above normal operating pressure **P1** and into overload pressure **P2**. The increase to overload pressure **P2** triggers pressure relief valve **252** to allow a flow of liquid from liquid chamber **212** into circulation

tube **254** and through joint **256** and channel **258** into mixed fluid chamber **214**, as indicated by arrows **262**. After the hard landing, or overload, event, the pressure in shock strut **202** may decrease below overload pressure **P2** back to normal operating pressure **P1**. However, pressure relief valves **252** do not allow a flow of liquid from circulation tube **254** back to lower chamber.

(22) Referring now to FIGS. **4A** and **4B**, cross section views of a metering pin **400** and a metering pin **450** that may be used in shock strut **202** are illustrated, in accordance with various embodiments. Referring first to FIG. **4A**, metering pin **400** has a first end **402** including a first opening **404**, a second end **406** including a second opening **408**, and a channel **410** formed through metering pin **400** extending from first opening **404** to second opening **408**. Channel **410** may be an example of channel **258** described above in FIGS. **3A** and **3B**. Channel **410** allows fluid to flow through metering pin **400**, as previously described.

(23) Referring next to FIG. **4B**, metering pin **450** has a first end **452** including a first opening **454**, a second end **456** including a second opening **458a** and a third opening **458b**, and a channel **460** formed through metering pin **450** extending from first opening **454** to second opening **458a** and third opening **458b**. Channel **460** may be an example of channel **258** described above in FIGS. **3A** and **3B**. Channel **460** allows fluid to flow through metering pin **400**, as previously described. Second opening **258a** and third opening **258b** may be formed through a sidewall of metering pin **450**, directing the flow of fluid perpendicular to channel **460** (e.g., along the x-axis). In various embodiments, second opening **258a** and third opening **258b** may be formed at an angle α_1 relative to metering pin **450** (e.g., the z-axis). In the illustrated embodiment, angle α_1 is about 90° from metering pin **450** resulting in a fluid flow that is perpendicular to channel **460**. In various embodiments, second opening **258a** and third opening **258b** may be formed at a downward angle (e.g., in the negative z-direction) or at an upward angle (e.g., in the positive z-direction) relative to metering pin **450**. In various embodiments, angle α_1 is about 30° to about 150° from metering pin **450** (e.g., along the z-axis), and more specifically about 45° to about 135° . While two openings are illustrated (i.e., second opening **258a** and third opening **258b**), in various embodiments, a plurality of openings may be formed through the sidewall of metering pin **450**. In various embodiments, the plurality of openings may be formed circumferentially around metering pin **450** and/or along a length of metering pin **450** (e.g., along the z-axis). In various embodiments, this reduces spray of fluid in mixed fluid chamber **214** of shock strut **202**. In various embodiments, a metering pin may include a features of metering pin **400** and metering pin **450** such that a plurality of openings are formed through a sidewall of the metering pin and an opening is formed through a top end (e.g., in the positive z-direction) of the metering pin.

(24) Referring now to FIG. **5**, a plot **500** of a shock strut load **502** as a function of the shock strut stroke **504** is illustrated, in accordance with various embodiments. Plot **500** may be used to determine a calibration setting for pressure relief valve **252**. As previously mentioned, each shock strut **202** may have a different calibration setting for pressure relief valve **252**. A first line **506** illustrates a static shock strut load **502** increasing as the shock strut stroke **504** increases. Generally, as the shock strut stroke **504** of shock strut **202** increases (e.g., along axis of shock strut stroke **504**) the load **502** on shock strut **202** increases in a predictable manner, as illustrated by first line **506**. The slope of first line **506** may be different for different shock struts **202**. A second line **508** illustrates a dynamic shock strut load **502** as with respect to the increase in shock strut stroke **504**. That is, second line **508** illustrates the shock strut load **502** on shock strut **202** during a landing event (or a dynamic event). As illustrated, the shock strut load **502** may increase dramatically with little to no increase in the stroke of shock strut **202**. Again, each shock strut design (e.g., shock strut **202**) may have different dynamic characteristics identified by second line **508**. Plot **500** further includes a threshold **510** indicating a threshold at which normal operating pressure **P1** becomes overload pressure **P2**. A hard landing, or overload, event may be defined as a specific shock strut load **502** for each shock strut by using second line **508** and threshold **510**. The calibration settings for pressure relief valve **252** may be determined for each different shock strut based on what load

defines a hard landing event as provided by second line **508**. In various embodiments, threshold **510** may be based on information from plot **500**.

(25) Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Different cross-hatching is used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

(26) Systems, methods, and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “various embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(27) Numbers, percentages, or other values stated herein are intended to include that value, and also other values that are about or approximately equal to the stated value, as would be appreciated by one of ordinary skill in the art encompassed by various embodiments of the present disclosure. A stated value should therefore be interpreted broadly enough to encompass values that are at least close enough to the stated value to perform a desired function or achieve a desired result. The stated values include at least the variation to be expected in a suitable industrial process, and may include values that are within 5% of a stated value. Additionally, the terms “substantially,” “about” or “approximately” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, the term “substantially,” “about” or “approximately” may refer to an amount that is within 5% of a stated amount or value.

(28) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112 (f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(29) Finally, it should be understood that any of the above-described concepts can be used alone or in combination with any or all of the other above-described concepts. Although various embodiments have been disclosed and described, one of ordinary skill in this art would recognize that certain modifications would come within the scope of this disclosure. Accordingly, the

description is not intended to be exhaustive or to limit the principles described or illustrated herein to any precise form. Many modifications and variations are possible in light of the above teaching.

Claims

1. A shock strut assembly, comprising: a mixed fluid chamber; a liquid chamber coupled to the mixed fluid chamber; a metering pin having a first end and a second end and a channel formed therein extending from the first end to the second end, the first end disposed in the liquid chamber and the second disposed in the mixed fluid chamber, wherein the metering pin is tapered with the first end of the metering pin being wider than the second end of the metering pin and wherein the metering pin includes a plurality of metering flutes coupled to the channel and disposed through a sidewall of the metering pin along a length of the metering pin; an orifice plate configured to separate the liquid chamber from the mixed fluid chamber, the orifice plate comprising a metering pin aperture configured to receive the second end of the metering pin, wherein the plurality of metering flutes defines a flow channel between the metering pin and the metering pin aperture of the orifice plate through which liquid may travel from the liquid chamber to the mixed fluid chamber in response to a shock strut moving to a compressed position from an extended position; and a pressure relief valve fluidly coupled to the liquid chamber and the channel in the metering pin, the pressure relief valve configured to allow a flow of a fluid from the liquid chamber to the channel in the metering pin.
2. The shock strut assembly of claim 1, further comprising: a circulation tube fluidly coupled to the pressure relief valve; and a joint fluidly coupled to the channel and the circulation tube.
3. The shock strut assembly of claim 1, wherein the pressure relief valve is configured to prevent the flow of the fluid from the channel to the liquid chamber.
4. The shock strut assembly of claim 1, wherein the second end of the metering pin includes an opening to the channel along an axis of the metering pin.
5. The shock strut assembly of claim 1, wherein the second first end of the metering pin includes a first opening to the channel and a second opening to the channel that are perpendicular to an axis of the metering pin.
6. The shock strut assembly of claim 1, wherein the pressure relief valve is configured to allow the flow of the fluid in response to a pressure in the shock strut assembly being above a threshold.
7. The shock strut assembly of claim 6, wherein the pressure relief valve is configured to prevent the flow of the fluid in response to the pressure in the shock strut assembly being below the threshold.
8. A shock strut assembly, comprising: a shock strut cylinder defining a mixed fluid chamber; a shock strut piston defining a liquid chamber, the shock strut piston operatively coupled to slide within the shock strut cylinder; a metering pin having a first end and a second end and configured to extend from the shock strut piston into the shock strut cylinder, wherein the metering pin is tapered with the first end of the metering pin being wider than the second end of the metering pin and wherein the metering pin includes a plurality of metering flutes coupled to the channel and disposed through a sidewall of the metering pin along a length of the metering pin; an orifice plate configured to separate the liquid chamber from the mixed fluid chamber, the orifice plate comprising a metering pin aperture configured to receive the second end of the metering pin, wherein the plurality of metering flutes defines a flow channel between the metering pin and the metering pin aperture of the orifice plate through which a liquid may travel from the liquid chamber to the mixed fluid chamber in response to a shock strut moving to a compressed position from an extended position; a channel formed within the metering pin from the first end to the second end, the channel configured to allow a flow of the liquid from the liquid chamber to the mixed fluid chamber; and a pressure relief valve fluidly coupled to the liquid chamber and the channel.

9. The shock strut assembly of claim 8, further comprising: a circulation tube fluidly coupled to the pressure relief valve; and a joint fluidly coupled to the channel and the circulation tube.
 10. The shock strut assembly of claim 8, wherein the pressure relief valve is configured to allow the flow of the liquid in response to a pressure in the shock strut assembly being above a threshold.
 11. The shock strut assembly of claim 8, wherein the pressure relief valve is configured to prevent the flow of the liquid in response to a pressure in the shock strut assembly being below a threshold.
 12. The shock strut assembly of claim 8, wherein the pressure relief valve is configured to prevent the flow of the liquid from the channel to the liquid chamber.
 13. The shock strut assembly of claim 8, wherein the shock strut piston is configured to slide in a first direction and wherein the second end of the metering pin includes an opening to the channel that is along the first direction.
 14. The shock strut assembly of claim 8, wherein the shock strut piston is configured to slide in a first direction and the first end of the metering pin includes a first opening to the channel and a second opening to the channel that are perpendicular to an axis of the metering pin.
 15. A landing gear assembly, comprising: a shock strut; a wheel assembly coupled to the shock strut; and an energy absorption system coupled to the shock strut, the energy absorption system including: a shock strut cylinder defining a mixed fluid chamber; a shock strut piston defining a liquid chamber, the shock strut piston operatively coupled to slide within the shock strut cylinder; a metering pin having a first end and a second end, the first end disposed in the shock strut piston and the second end disposed in the shock strut cylinder, wherein the metering pin is tapered with the first end of the metering pin being wider than the second end of the metering pin and wherein the metering pin includes a plurality of metering flutes coupled to the channel and disposed through a sidewall of the metering pin along a length of the metering pin; an orifice plate configured to separate the liquid chamber from the mixed fluid chamber, the orifice plate comprising a metering pin aperture configured to receive the second end of the metering pin, wherein the plurality of metering flutes defines a flow channel between the metering pin and the metering pin aperture of the orifice plate through which liquid may travel from the liquid chamber to the mixed fluid chamber in response to the shock strut moving to a compressed position from an extended position; a channel formed along an inner diameter of the metering pin from the first end to the second end; and a pressure relief valve coupled to the liquid chamber and the channel.
 16. The landing gear assembly of claim 15, the energy absorption system further comprising: a circulation tube fluidly coupled to the pressure relief valve; and a joint fluidly coupled to the channel and the circulation tube.
 17. The landing gear assembly of claim 15, wherein the pressure relief valve is configured to allow a flow of a fluid in response to a pressure in the shock strut being above a threshold.
 18. The landing gear assembly of claim 15, wherein the pressure relief valve is configured to prevent a flow of a fluid in response to a pressure in the shock strut being below a threshold.
 19. The landing gear assembly of claim 15, wherein the pressure relief valve is configured to prevent a flow of a fluid from the channel to the liquid chamber.
 20. The landing gear assembly of claim 15, wherein the second first end of the metering pin includes a first opening to the channel and a second opening to the channel that are perpendicular to an axis of the metering pin.
-